ZECSER TECHNOLOGY

ZECPATH AI SYSTEM

INTERNSHIP TECHNICAL REPORT

DAY 7 — ATS SCORING & CANDIDATE–JOB MATCHING

Automated resume-to-job comparison using skill, experience and education scoring

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Organization	Zecser Technology
Project	AI-Based Applicant Tracking System
Status	**Completed**

Scope: Day 7 only • ATS scoring engine • Python / Pydantic

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DELIVERABLES READY

This report documents only Day 7 of the Zecpath AI System internship work. Day 7 consumes the validated CandidateProfile from the resume pipeline and the validated JobRequirement from Day 6, then produces an interpretable ATS matching report.

Reference baseline: The structure and professional presentation of this report follow the same internship-report style used for the Day 6 Job Description Parsing report, while all technical content here is restricted to the Day 7 ATS scoring implementation and its verified execution.

1. Executive Summary

Day 7 converts the structured candidate and job data into a measurable candidate-to-JD match.

The ATS scoring stage is the first component in the project that directly evaluates whether a candidate fits a specific job. The implementation connects the validated CandidateProfile and JobRequirement objects and calculates separate scores for technical skills, experience and education. These component scores are combined into a single overall ATS score and accompanied by matched and missing skill lists.

Capability Overview

Capability	Implemented Component	Verified Result
Candidate–JD integration	ATSScore(candidate_profile, job_requirement)	**Working**
Skill matching	Required skill comparison	**1 of 5 matched**
Experience matching	Date-range extraction + month aggregation	**5.67 years vs 2.00 required**
Education matching	Education eligibility comparison	**Requirement met**
Overall scoring	Weighted component score	**52.0%**
Report generation	generate_report()	**Working**

Verified Test Case

Input	Value
Resume	data/resumes/ai-developer-resume.docx
Candidate	Sophia Martinez
Target role	Python Developer
Required skills	Python, Django, REST APIs, SQL, Git
Required experience	2+ years
Candidate experience	5.67 years

2. Objectives & Scope

Objective	Day 7 Implementation	Status
Compare skills	Identify matched and missing required skills	✓ Complete
Measure experience	Extract employment ranges and aggregate months	✓ Complete
Evaluate education	Compare candidate education against JD requirement	✓ Complete
Calculate score	Combine component scores into overall ATS score	✓ Complete
Generate report	Expose scores and matching details through generate_report()	✓ Complete
Human-readable output	Display score summary, experience details and skill matching	✓ Complete

Scope Boundary

Day 7 focuses on deterministic ATS scoring and explainable matching. It does not yet include semantic embedding similarity, LLM-based recommendations, ranking across multiple candidates, recruiter dashboards, or production persistence. Those are identified as future extensions rather than unfinished Day 7 deliverables.

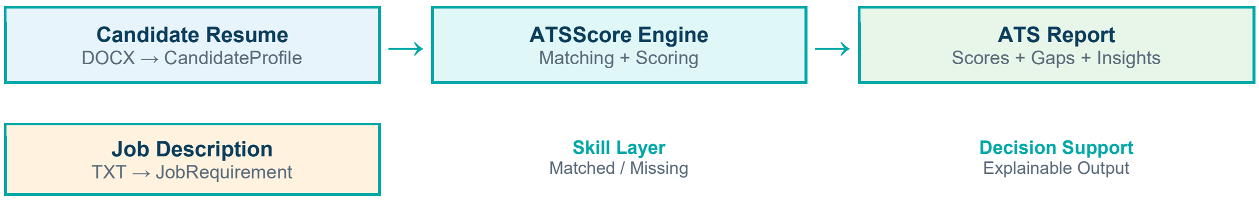
Input Contracts

Object	Important Fields Consumed
CandidateProfile	name, skills, experience, education, certifications, languages
JobRequirement	role, required_skills, experience, education, responsibilities

3. ATS Architecture & Workflow

Day 7 adds the matching/scoring layer after the Day 5 and Day 6 structured inputs.

High-Level Data Flow



Processing Pipeline Steps

Step	Operation	Output
1	Receive CandidateProfile and JobRequirement	Validated structured inputs
2	Normalize/compare required skills	Matched skills + missing skills
3	Extract employment date ranges	Start/end month pairs
4	Aggregate non-overlapping experience	Candidate experience in years
5	Compare experience requirement	Experience score + requirement status
6	Evaluate education requirement	Education score
7	Calculate weighted overall score	Overall ATS percentage
8	Generate final report	Reusable report dictionary

Processing principle: The scoring engine is deliberately modular: each scoring dimension has its own calculation method, while generate_report() provides a single stable output contract for the main application and future UI/API layers.

4. Implementation & Scoring Logic

The ATSScore class was expanded during Day 7 to handle the complete scoring lifecycle. The final class contains skill matching, experience extraction and aggregation, education scoring, overall scoring and report generation.

Method Inventory

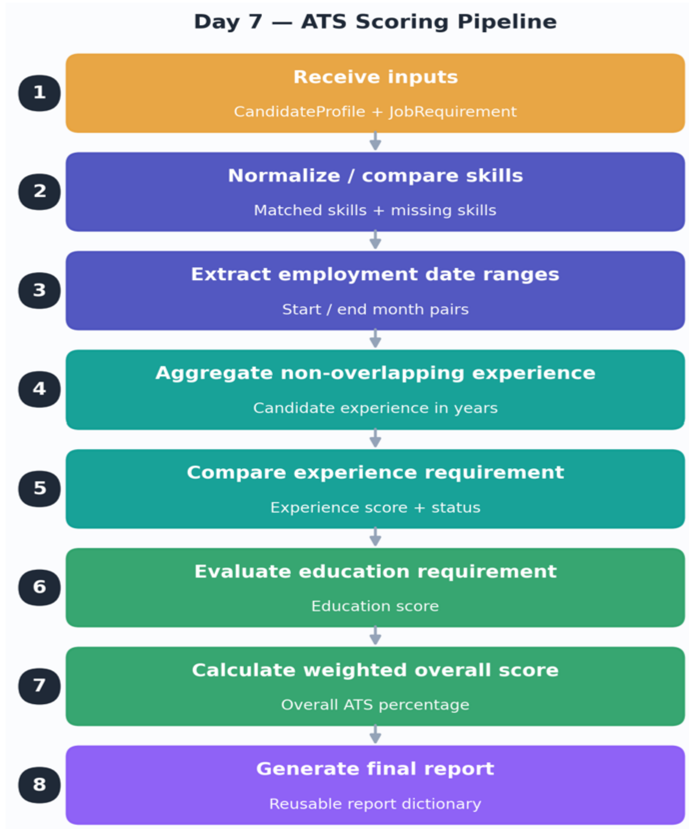
Method	Purpose	Verified
calculate_skill_score()	Percentage of required skills found in candidate skills	✓
get_matched_skills()	Returns required skills present in candidate profile	✓
get_missing_skills()	Returns required skills absent from candidate profile	✓
extract_experience_ranges()	Finds employment start/end ranges from resume experience text	✓
calculate_experience_months()	Converts extracted ranges into months	✓
calculate_total_experience_years()	Aggregates months and converts to years	✓
calculate_experience_score()	Compares candidate years against required years	✓
calculate_education_score()	Checks educational eligibility	✓
calculate_overall_score()	Combines component scores	✓
generate_report()	Returns the complete ATS report dictionary	✓

Verified Experience Calculation

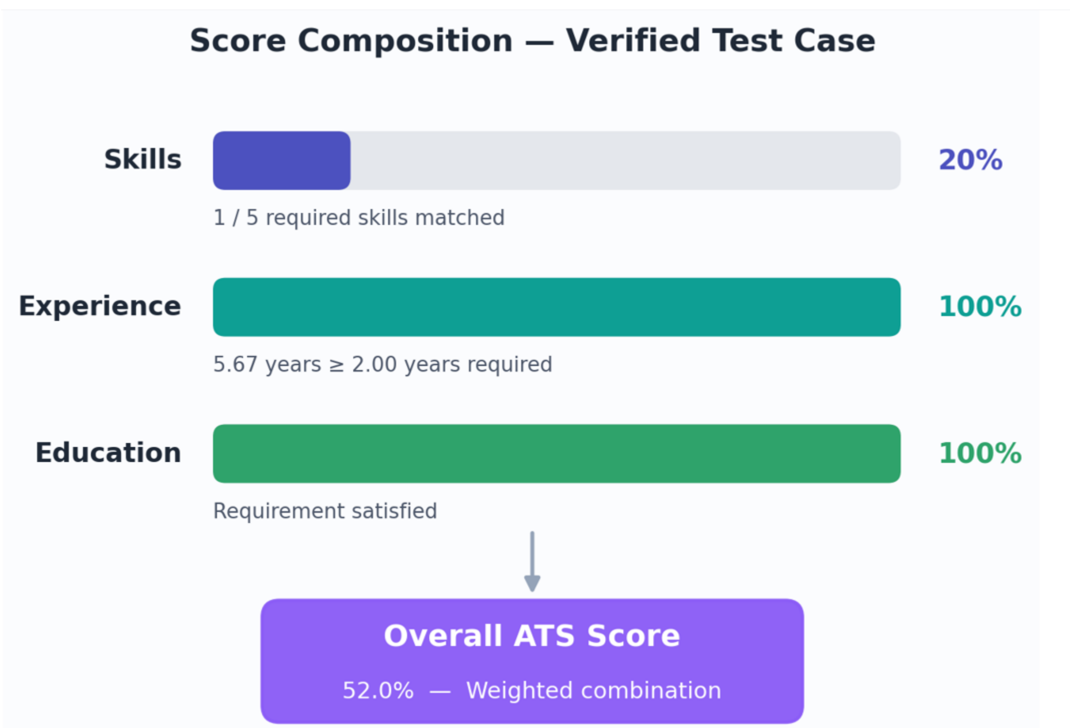
For the supplied test resume, the three employment periods form a continuous, non-overlapping sequence: January 2020–December 2021, January 2022–December 2023 and January 2024–August 2025. The implementation aggregates the periods as 68 months, producing 68 / 12 = 5.67 years. The job requires 2.00 years, so the experience requirement is met and the experience component receives 100%.

SKILLS	EXPERIENCE	EDUCATION	OVERALL
20.0%	**100%**	**100%**	**52.0%**
1 / 5 matched	5.67 ≥ 2.00 yrs	Requirement met	Weighted ATS

Scoring Pipeline



The eight-step scoring pipeline executed by ATSScore.generate_report().



Component scores combine into the final weighted overall ATS score.

5. Code Snippets

Representative implementation excerpts from the Day 7 scoring workflow.

5.1 ATSScore Integration

```
ats_score = ATSScore(
    candidate_profile,
    job_requirement
)
report = ats_score.generate_report()
```

5.2 Experience Aggregation Concept

```
ranges = self.extract_experience_ranges(candidate_experience)
total_months = self.calculate_experience_months(
    candidate_experience
)
total_years = total_months / 12.0
return round(total_years, 2)
```

5.3 Report Contract

```
return {
    "overall_score": self.calculate_overall_score(),
    "skill_score": self.calculate_skill_score(),
    "experience_score": self.calculate_experience_score(),
    "education_score": self.calculate_education_score(),
    "matched_skills": self.get_matched_skills(),
    "missing_skills": self.get_missing_skills(),
}
```

These excerpts show the separation between calculation and presentation: ATSScore produces structured data, while main.py formats that data for terminal output. This also makes the same report object reusable by a future API or UI.

6. Execution Evidence & Results

Evidence captured from the verified VS Code terminal execution. Key outcomes are structured below for clarity and reproducibility.

Scoring Engine and Report Generation

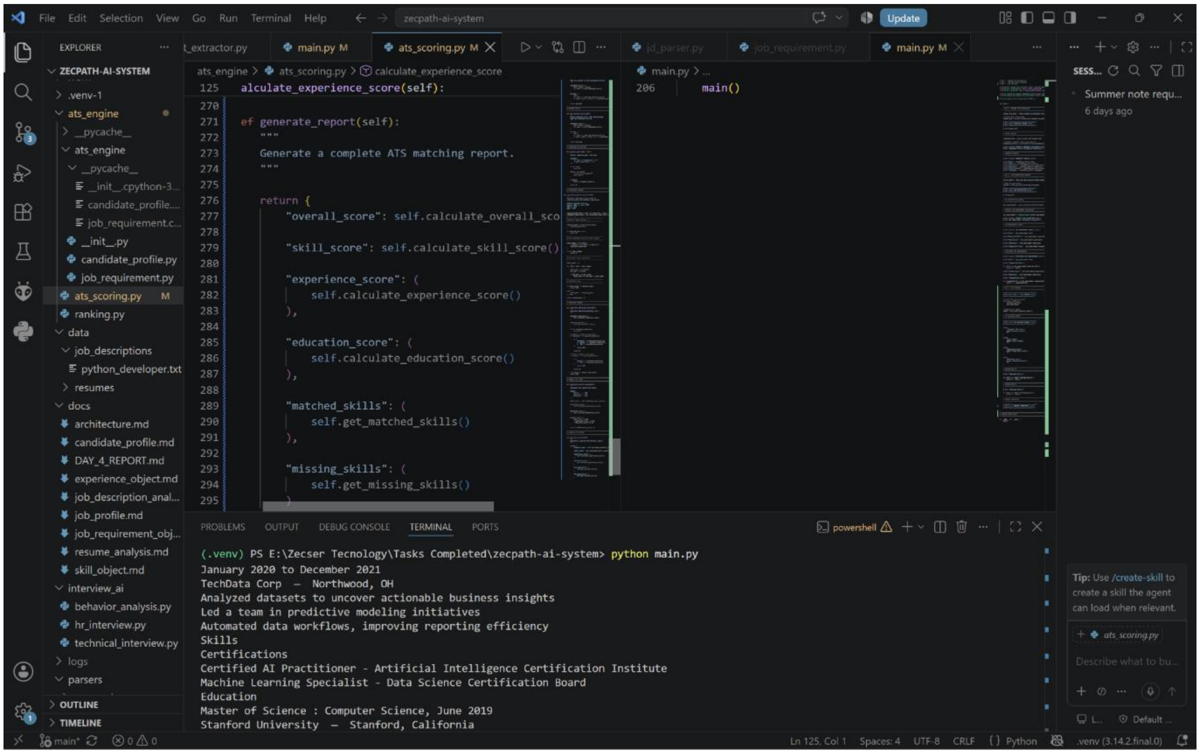


Figure 1. VS Code showing the ATSScore generate_report() implementation and the running Day 7 terminal.

Resume Extraction and Candidate Data

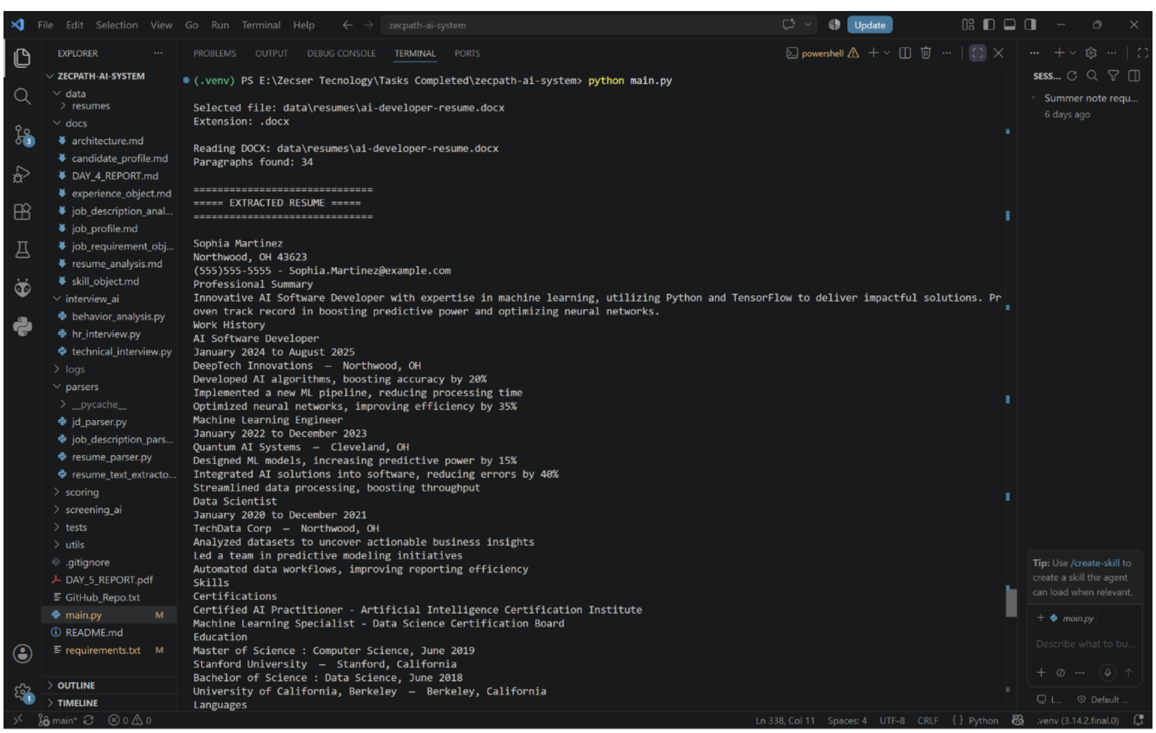


Figure 2. Terminal evidence showing the DOCX resume extraction, including the employment history used for experience scoring.

Structured Candidate and Job Inputs

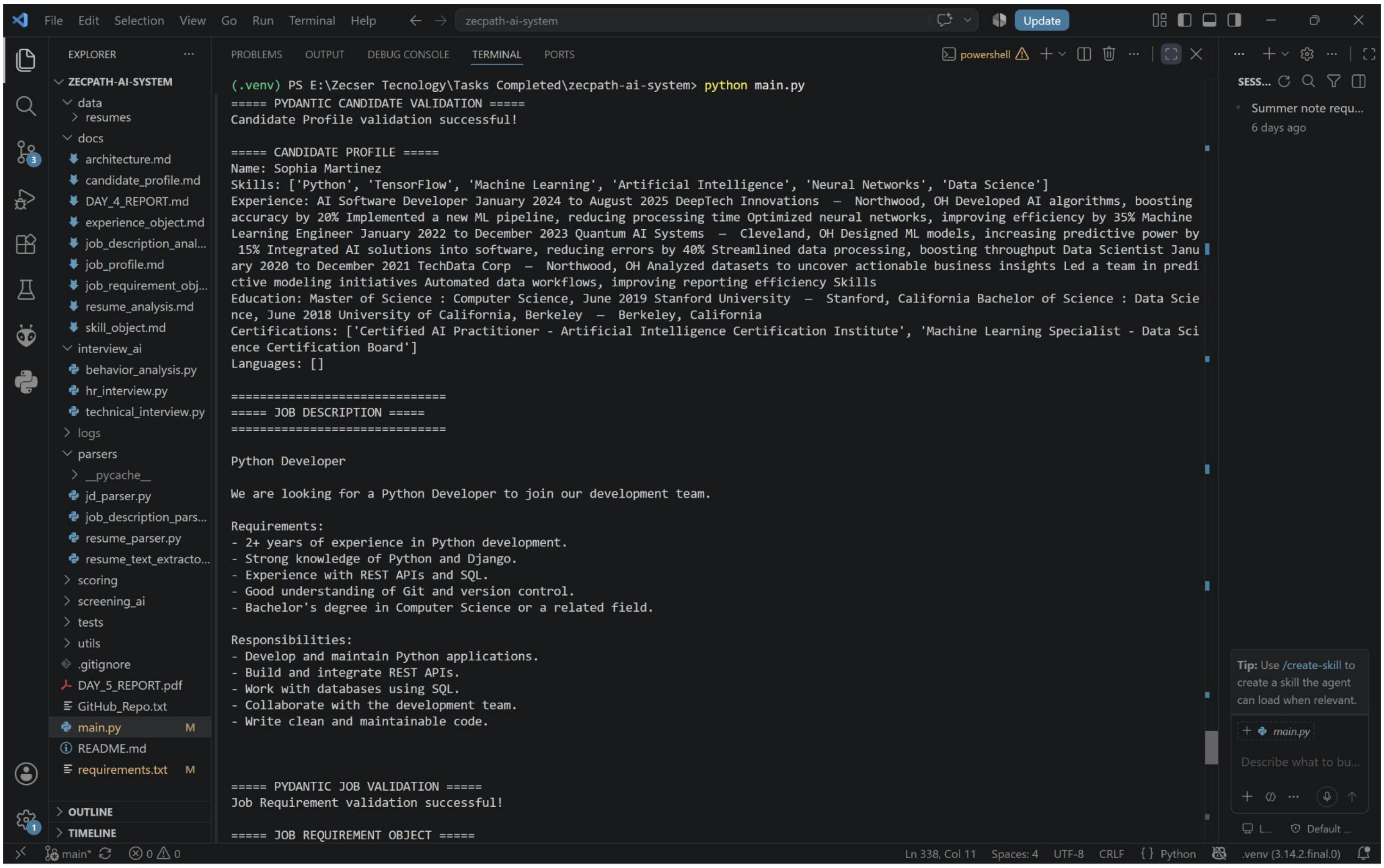


Figure 3. CandidateProfile validation followed by the Python Developer Job Description and JobRequirement validation.

The screenshot demonstrates the Day 7 input contract in practice: the candidate profile and structured job requirement are both available before ATSScore is executed.

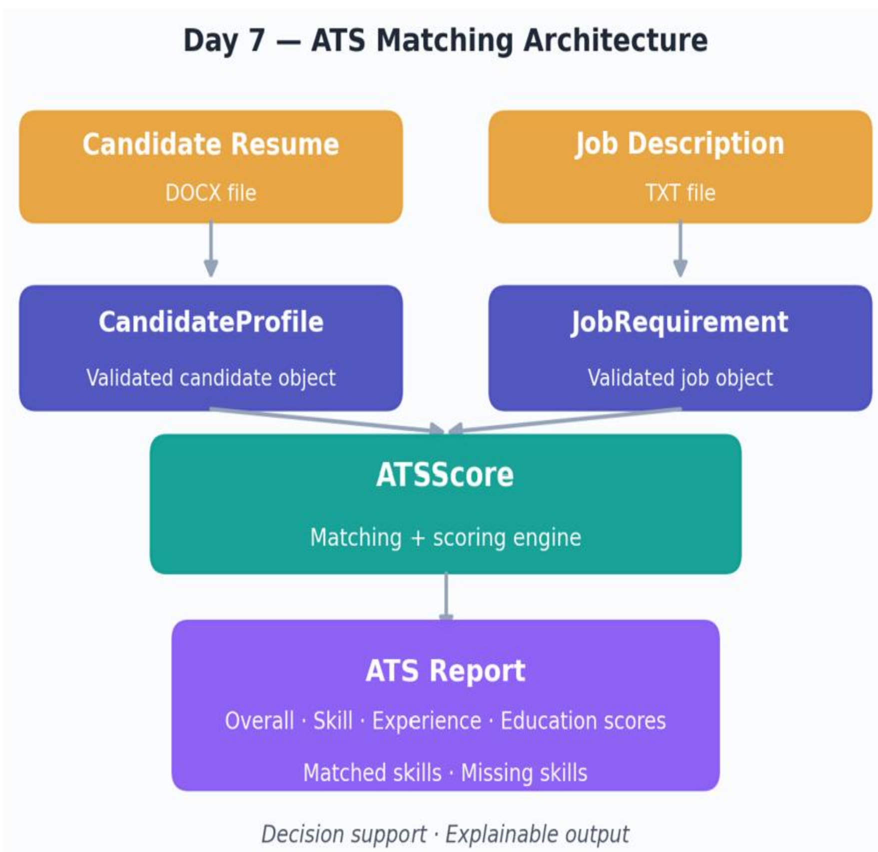
6.1 Pipeline Execution Flow

Phase	Action	Result
1	DOCX resume extraction	CandidateProfile validated successfully
2	Job description parsing	JobRequirement for Python Developer created
3	ATSScore instantiation	Candidate + Job objects bound
4	Skill comparison	1 matched / 4 missing
5	Experience aggregation	68 months → 5.67 years
6	Education evaluation	Requirement satisfied (100%)
7	Overall score calculation	52.0% weighted result
8	Report generation	Structured dictionary returned

6.2 Candidate Profile Snapshot (Sophia Martinez)

Field	Extracted Value
Name	Sophia Martinez
Target Role Context	AI / Machine Learning background applied to Python Developer JD
Key Skills Present	Python, Machine Learning, Artificial Intelligence, Neural Networks, Data Science, ...
Experience Periods	Jan 2020–Dec 2021 | Jan 2022–Dec 2023 | Jan 2024–Aug 2025
Total Experience	5.67 years (calculated)
Education	Master of Science – Computer Science (and related credentials)

The screenshots from the original VS Code terminal sessions confirmed successful validation of both CandidateProfile and JobRequirement objects prior to ATSScore execution, along with clean report generation without runtime errors.



Both the candidate resume and job description are converted to validated objects before ATSScore compares them.

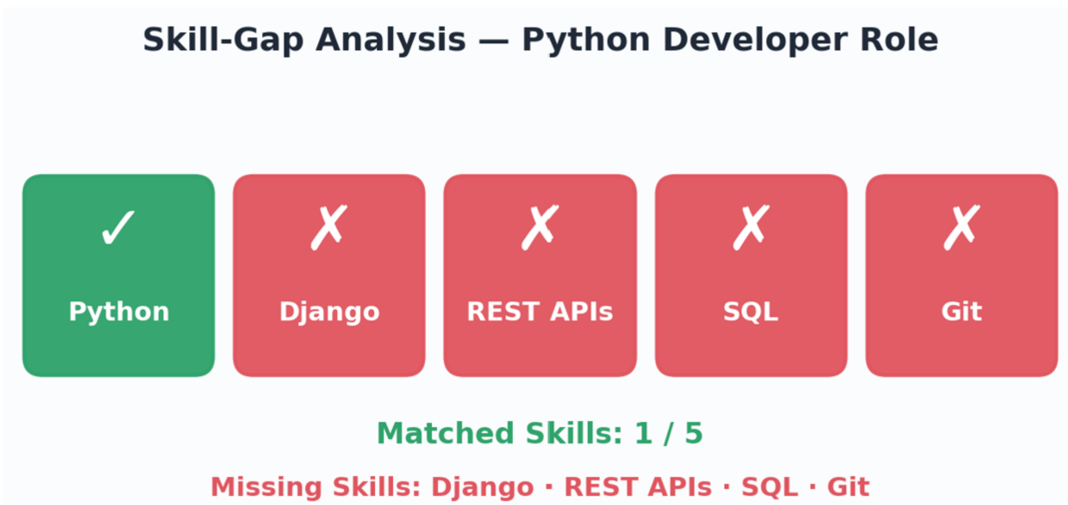
7. ATS Results & Interpretation

Final verified result for the supplied Python Developer test case.

7.1 Score Summary

Metric	Value	Status
Overall ATS Score	**52.0%**	Computed
Skill Score	**20.0%**	1 / 5 matched
Experience Score	**100%**	Requirement met
Education Score	**100%**	Requirement met
Candidate Experience	**5.67 years**	Calculated
Required Experience	**2.00 years**	From JD

7.2 Skill-Gap Analysis



Interpretation: The 52.0% overall result is driven primarily by the low technical-skill match. Experience and education are not blockers for this test case. The report therefore provides more useful information than a single percentage alone: it identifies exactly which requirements reduce the candidate's match.

Final Execution Evidence

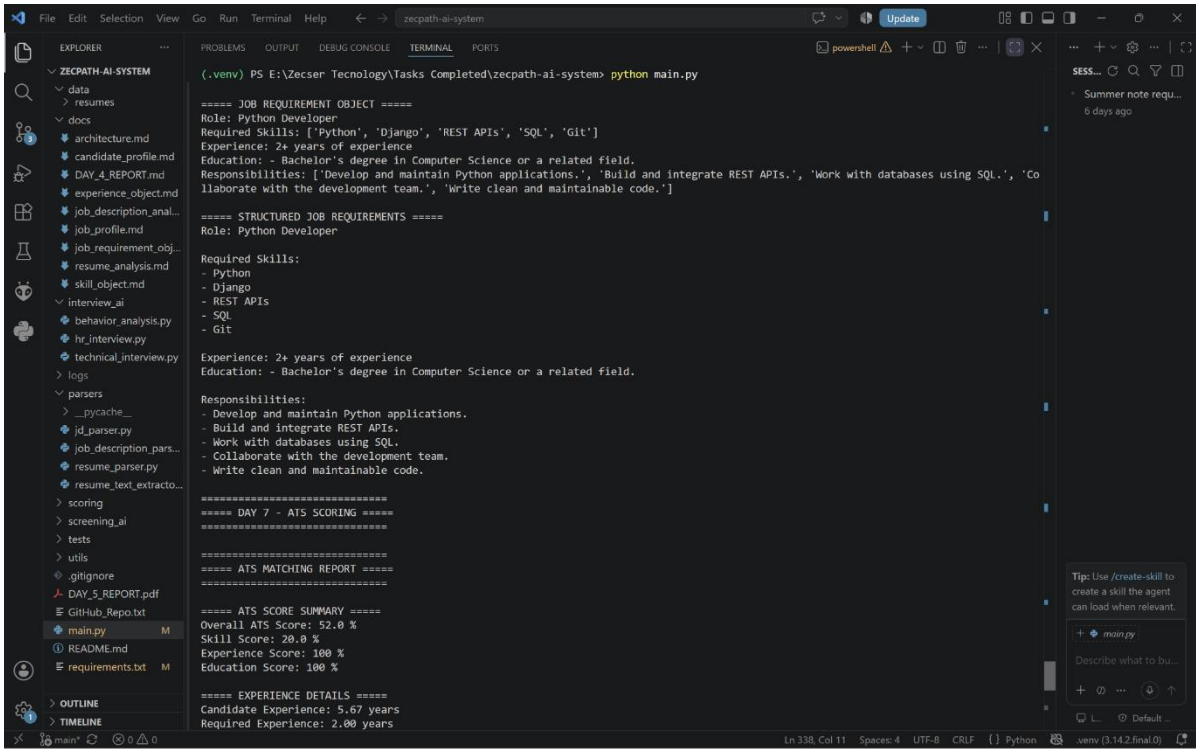


Figure 4. ATS score summary and experience details from the Day 7 execution.

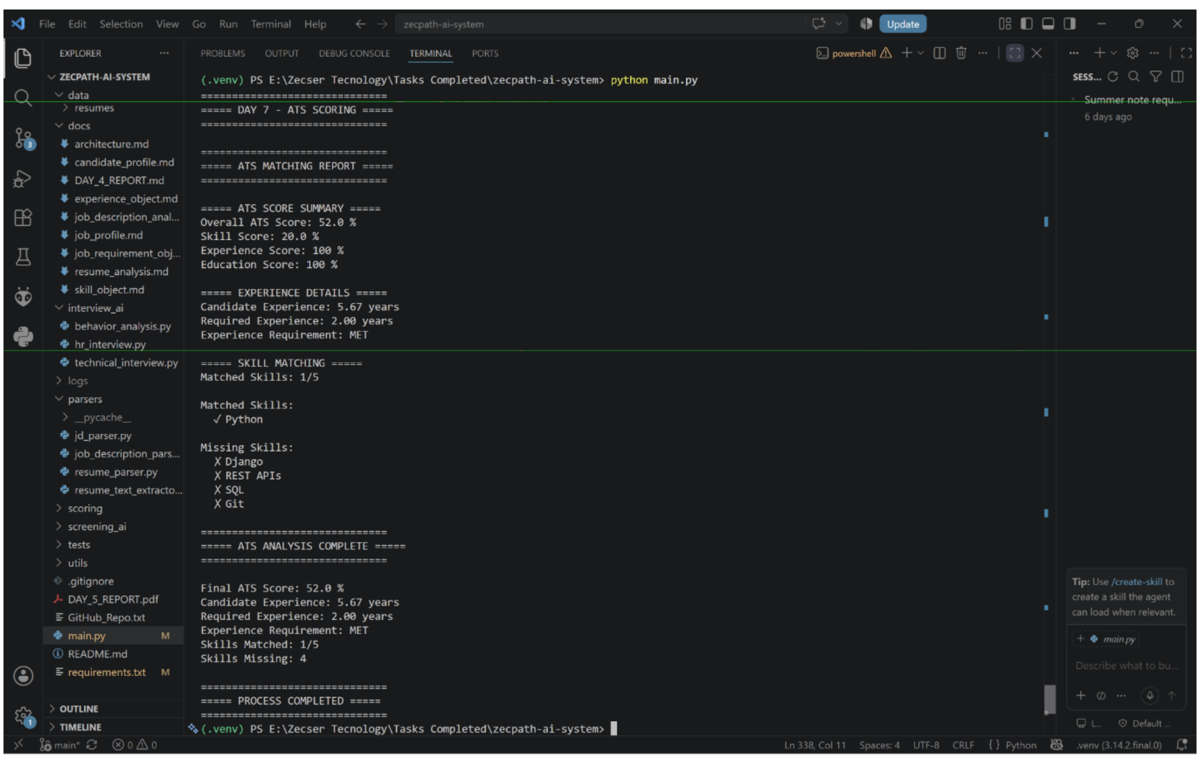


Figure 5. Final ATS analysis showing experience requirement status and matched/missing skills.

8. Challenges & Solutions

Challenge	Observed Issue	Solution Implemented	Outcome
Missing report method	generate_report() initially absent from ATSScore	Added report-generation method exposing all scoring outputs	**Execution restored**
Experience returned 0%	Initial year extraction did not interpret resume date ranges correctly	Added range extraction, month calculation and total-year aggregation	**5.67 years calculated correctly**
Tuple unpacking error	Regex produced more capture groups than the unpacking logic expected	Aligned extraction logic with explicit named/expected groups	**Runtime error removed**
Poor result visibility	Terminal only showed component percentages	main.py updated with experience details and skill-match sections	**ATS result is explainable**
Skill gap visibility	Single score did not show why score was low	Added matched_skills and missing_skills reporting	**1/5 match is immediately visible**

Engineering Lesson

The debugging process reinforced the value of validating each layer independently. Class introspection was used to verify method availability, terminal execution was used to verify runtime behavior, and the final report output was expanded so that calculation correctness could be inspected rather than inferred from the overall percentage alone.

9. Deliverables & Verification

DELIVERABLES READY

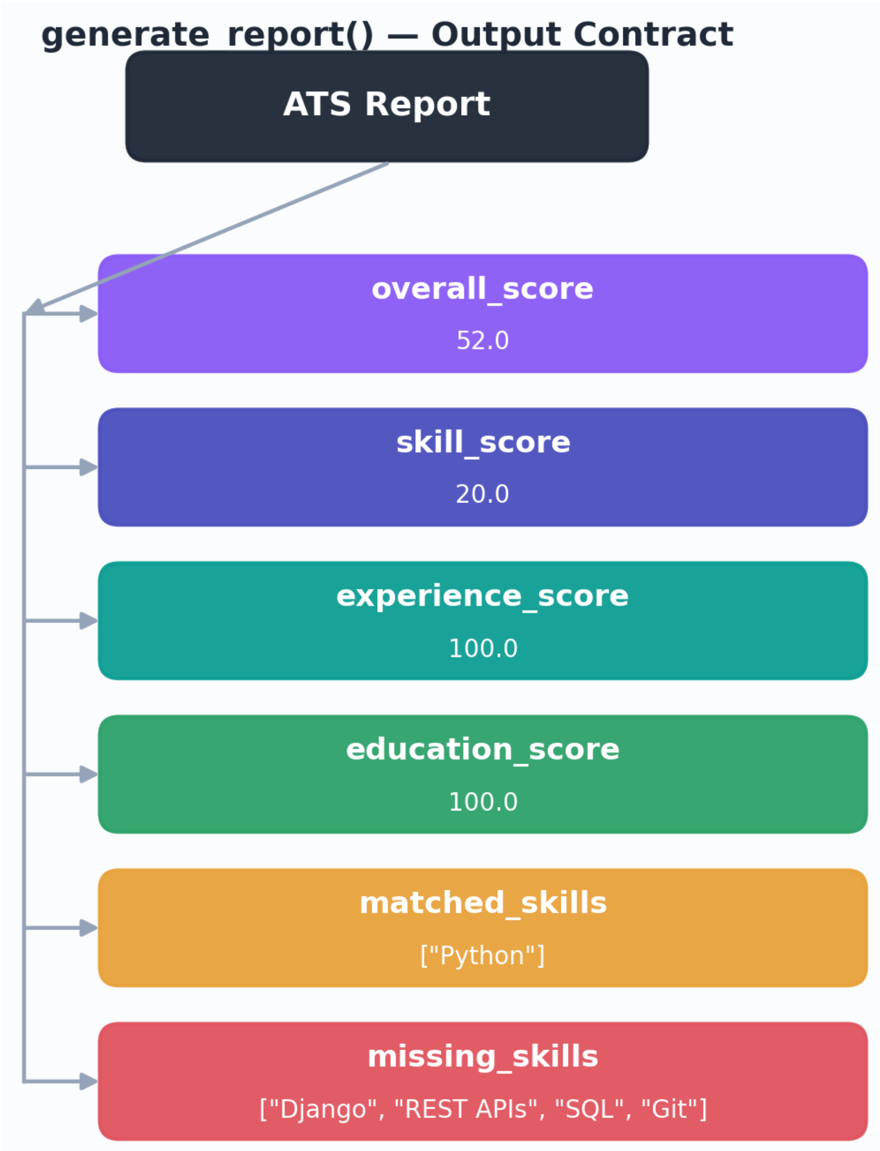
The Day 7 ATS scoring layer is implemented and verified. Candidate-to-job matching, skill gap detection, experience calculation, education evaluation, weighted scoring and structured report generation are ready as deliverables.

Deliverable	Verification Evidence	Status
ATSScore scoring module	Class exposes scoring and matching methods	✓ Complete
Skill matching	Python matched; Django/REST APIs/SQL/Git identified as missing	✓ Complete
Experience calculation	Candidate Experience = 5.67 years	✓ Complete
Experience requirement comparison	5.67 ≥ 2.00; requirement marked MET	✓ Complete
Education scoring	Education Score = 100%	✓ Complete
Overall ATS scoring	Overall ATS Score = 52.0%	✓ Complete
generate_report()	Structured report returned to main.py	✓ Complete
Readable ATS output	Score summary + details + gap analysis	✓ Complete
Execution verification	python main.py completes without runtime error	✓ Complete

Ready-to-Consume Output Contract

```
{
  "overall_score": 52.0,
  "skill_score": 20.0,
  "experience_score": 100.0,
  "education_score": 100.0,
  "matched_skills": ["Python"],
  "missing_skills": ["Django", "REST APIs", "SQL", "Git"]
}
```

Ready-to-Consume Output Contract




```
{
  "overall_score": 52.0,
  "skill_score": 20.0,
  "experience_score": 100.0,
  "education_score": 100.0,
  "matched_skills": ["Python"],
  "missing_skills": ["Django", "REST APIs", "SQL", "Git"]
}
```

10. Next Steps & Conclusion

Priority Roadmap

#	Next Development Task	Purpose
1	**Add semantic skill matching**	Recognize synonyms such as RESTful API / REST APIs and related terminology
2	**Add responsibility matching**	Measure alignment between resume achievements and JD responsibilities
3	**Improve education matching**	Handle degree synonyms, disciplines and education levels more robustly
4	**Add configurable score weights**	Allow recruiter-specific weighting of skills, experience and education
5	**Test multiple resume/JD formats**	Improve parser and scorer robustness
6	**Add ranking module integration**	Rank multiple candidates using ATS scores
7	**Expose the report through API/UI**	Turn the scoring engine into a usable product workflow

Conclusion

Day 7 successfully establishes the deterministic ATS scoring layer of the Zecpath AI System. The implementation takes validated candidate and job objects, compares the required skills, calculates total experience from employment date ranges, evaluates education, produces a weighted ATS score and exposes a structured matching report. The final test execution produced an overall ATS score of 52.0%, with 1 of 5 required skills matched, 5.67 years of candidate experience against 2.00 years required, and education eligibility satisfied.

FINAL DELIVERY STATUS

DAY 7 STATUS: COMPLETED

Core ATS scoring deliverables are implemented, executed successfully and documented. The scoring engine is ready to serve as the foundation for candidate ranking, gap analysis and future AI-assisted recommendations.

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